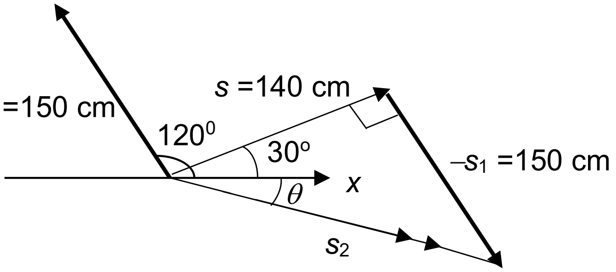
Units of $\sqrt{g\lambda} = \sqrt{m^2\,s^{-2}} = m\,s^{-1}$

**2** **A**




$s_2 = s - s_1$

By Pythagoras theorem, $s_2 = \sqrt{140^2 + 150^2} = 205$ cm

$\tan\theta = \dfrac{150}{140}$ or $\theta = 47^o$

Hence $\theta = -17^o$ anti-clockwise with respect to positive x-axis

**3** **B** $a = (v - u)/t = (0 - 15)/1.2 = -12.5\ m\ s^{-2}$